Problem 2

$$\vec{j} = \sigma(\omega)\vec{E} = \frac{\sigma_0}{1-i\omega\tau}\vec{E} \qquad \text{(Ohm's law)}$$

$$\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \qquad \text{(Gauss' law)}$$

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{j} = 0 \Rightarrow i\omega\rho = \nabla\cdot\vec{j} \quad \text{(continuity of charge flow)}$$

Excluding ρ with the help of the last equation $\rho = -\frac{i\nabla\cdot j}{\omega}$ and using it in the second one:

$$\nabla\cdot\vec{E} = -\frac{i}{\varepsilon_0\omega}\nabla\cdot\vec{j} \qquad (*)$$

On the other hand taking $\nabla\cdot$ operation to both sides of the Ohm's law we get

$$\nabla\cdot\vec{j} = -\frac{\sigma_0}{1-i\omega\tau}\nabla\cdot\vec{E} \qquad (**)$$

These two last equations are compatible only if

$$\frac{\sigma_0}{1-i\omega\tau} \times \frac{i}{\omega\varepsilon_0} = -1$$

This is a quadratic equation:

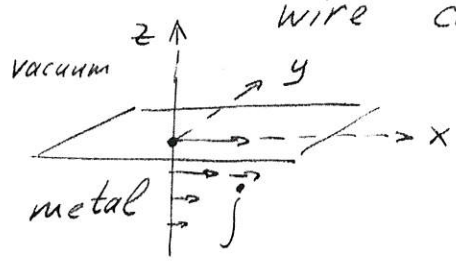
$$\omega\varepsilon_0 - i\omega^2\tau\varepsilon_0 = i\sigma_0 = -i\varepsilon_0\omega_p^2\tau$$

or $\boxed{\omega^2 = \omega_p^2 - \frac{i\omega}{\tau}}$ In the limit of $\omega_p\tau \gg 1$ the last term is small in which ω can be replaced with ω_p to the lowest order:

$$\omega^2 = \omega_p^2 - \frac{i\omega_p}{\tau} \Rightarrow \omega = \sqrt{\omega_p^2 - \frac{i\omega_p}{\tau}} \approx \omega_p - \frac{i}{2\tau}$$

$$\omega' = \omega_p \quad \text{and} \quad \omega'' = -1/2\tau \ll \omega'$$

Problem 2 when $\delta \ll a$ the surface of the wire can be approximated as flat



$\vec{j} = \sigma_0 \vec{E}$

both $\vec{j}$ and $\vec{E}$ are along x.

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = i\omega \vec{B}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j} = \mu_0 \sigma_0 \vec{E}$$ (displacement current is neglected as a small correction at $\omega \ll \omega_{pl}$)

$$\nabla \times (\nabla \times \vec{E}) = i\omega \mu_0 \sigma_0 \vec{E}$$

Since $\nabla \cdot \vec{E} = 0$ (why?) $\quad \nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$

$$\nabla^2 \vec{E} = -i\omega \mu_0 \sigma_0 \vec{E}$$

In our setup $\vec{E} = (E(z), 0, 0)$

$$\frac{d^2 E}{dz^2} = -i\omega \mu_0 \sigma_0 E$$ The solution, decaying when $z \to -\infty$ is

$$E(z) = E_0 e^{\sqrt{-i\omega\mu_0\sigma_0}\, z} = E_0 e^{\frac{z}{\delta}} e^{-\frac{iz}{\delta}}, \quad \delta = \sqrt{\frac{2}{\omega \mu_0 \sigma_0}}$$

as $\sqrt{-i} = \frac{-1-i}{\sqrt{2}}, \frac{1-i}{\sqrt{2}} \Rightarrow$ the second value should be chosen

We can now write actual electric field as the real part of the above solution:

$$E(z,t) = E_0 \, \mathrm{Re}\, e^{z/\delta} e^{-iz/\delta} e^{-i\omega t} =$$

$$= \rho_0 j_0 e^{z/\delta} \cos(\omega t + z/\delta), \quad \text{where } \rho_0 = \frac{1}{\sigma_0}$$

we introduced the amplitude of current j_0 at

the surface. It can be related to the total current via

$$I_0 = 2\pi a j_0 \int_{-\infty}^{0} dz\, e^{z/\delta} \cos\left(\frac{z}{\delta}\right) =$$

$$= 2\pi a j_0 \, \mathrm{Re} \int_{-\infty}^{0} dz\, e^{\frac{z}{\delta}(1-i)} =$$

$$= 2\pi a j_0 \, \mathrm{Re} \left(\frac{\delta}{1-i}\right) = \pi \delta a \quad \Rightarrow \boxed{j_0 = \frac{I_0}{\pi a \delta}}$$

Thus,

$$E(z,t) = \rho_0 \frac{I_0}{\pi a \delta} e^{z/\delta} \cos(\omega t + z/\delta)$$

Total Joule heat: per unit length:

$$Q = 2\pi a \int_{-\infty}^{0} dz\, \overline{E(z,t) \cdot j(z,t)} = \frac{2\pi a}{\rho_0} \int_{-\infty}^{0} dz \overline{\left(E(z,t)\right)^2}$$

$$= \frac{2\pi a}{\rho_0} \left(\frac{\rho_0 I_0}{\delta \pi a}\right)^2 \int_{-\infty}^{0} dz\, e^{2z/\delta} \overline{\cos^2(\omega t + z/\delta)}$$

The time average is simply $\overline{\cos^2(\omega t + z/\delta)} = \frac{1}{2}$

and $\int_{-\infty}^{0} dz\, e^{2z/\delta} = \frac{\delta}{2}$, so that

$$\boxed{Q = \frac{\rho_0 I_0^2}{\pi a \delta} = \frac{I_0^2}{\pi a \delta \sigma_0}}$$

P3 Static charge in plasma.

Investigate ~~distribution of~~ electrostatic potential $\varphi(\vec{r})$ around static charge Q in plasma. Use derived in class longitudinal dielectric function $\tilde{\varepsilon}^l$ (denoted ε_{zz} in the lecture note).

$$\vec{\nabla}\cdot\vec{D}(\vec{r},t) = \rho_{ext}(\vec{r},t) = Q\,\delta(\vec{r})$$

$$\Rightarrow \text{F.T.} \Rightarrow \quad i\vec{k}\cdot\vec{D}(\vec{k},\omega) = \rho_{ext}(\vec{k},\omega)$$

$$\vec{D}(\vec{k},\omega) = \underbrace{\tilde{\varepsilon}^l(\vec{k},\omega)}_{\varepsilon_{zz}(\vec{k},\omega)}\,\vec{E}(\vec{k},\omega)$$

But $\vec{E}(\vec{r},t) = -\vec{\nabla}\varphi(\vec{r},t) \Rightarrow \vec{E}(\vec{k},\omega) = -i\vec{k}\,\varphi(\vec{k},\omega)$

(since $\frac{\partial\vec{A}}{\partial t}=0$ for static field)

$\Rightarrow$ obtain $k^2\,\varepsilon_{zz}(k,\omega)\,\varphi(\vec{k},\omega) = \rho_{ext}(\vec{k},\omega)$

$$\varphi(\vec{k},\omega) = \frac{\rho_{ext}(\vec{k},\omega)}{k^2\,\varepsilon_{zz}(k,\omega)}$$

For a static point charge $\rho_{ext}(\vec{k},\omega) =$

$$= \int d^3\vec{r}\int_{-\infty}^{\infty} dt\; Q\,\delta(\vec{r})\,e^{i\omega t - i\vec{k}\cdot\vec{r}} = 2\pi Q\,\delta(\omega)$$

$$\Rightarrow \varphi(\vec{r},t) = \int\frac{d\omega}{2\pi}\int\frac{d^3\vec{k}}{(2\pi)^3}\,\varphi(\vec{k},\omega)\,e^{i\vec{k}\cdot\vec{r}-i\omega t} =$$

$$= \int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\int\frac{d^3\vec{k}}{(2\pi)^3}\,\frac{2\pi Q\,\delta(\omega)}{k^2\,\varepsilon_{zz}(k,\omega)}\,e^{i\vec{k}\cdot\vec{r}}\,e^{-i\omega t}$$

$$= Q\int\frac{d^3\vec{k}}{(2\pi)^3}\,\frac{e^{i\vec{k}\cdot\vec{r}}}{k^2\,\varepsilon_{zz}(k,\omega=0)}$$

From the lecture note

$$\varepsilon_{zz}(k,\omega) = \varepsilon_0 + \varepsilon_0 \frac{\omega_p^2}{(V_T k)^2}\left(1 - \Omega^2 + i\sqrt{\tfrac{\pi}{2}}\,\Omega\right) \quad \text{for } \Omega = \frac{\omega}{V_T k} \ll 1$$

$$\Rightarrow \varepsilon_{zz}(k,0) = \varepsilon_0\left(1 + \frac{\omega_p^2}{V_T^2 k^2}\right) \quad , V_T \equiv \sqrt{\frac{k_B T}{m}}$$

$$\omega_p = \sqrt{\frac{q^2 n}{\varepsilon_0 m}}$$

$$\Rightarrow k^2 \varepsilon_{zz}(k,0) = \varepsilon_0\left(k^2 + \frac{\omega_p^2}{V_T^2}\right)$$

$$\varphi(\vec{r},t) = \frac{Q}{\varepsilon_0}\int \frac{d^3\vec{k}}{(2\pi)^3}\,\frac{e^{i\vec{k}\cdot\vec{r}}}{k^2 + \varkappa^2}$$

where $\varkappa^2 = \dfrac{\omega_p^2}{V_T^2} = \dfrac{q^2 n}{\varepsilon_0 m}\,\dfrac{m}{k_B T} = \dfrac{q^2 n}{\varepsilon_0 k_B T}$

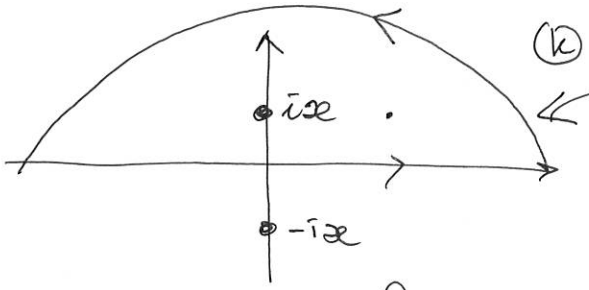
Calculate integral $\vec{k}\cdot\vec{r} = kr\cos\theta$

$$\Rightarrow I = \int_0^{2\pi}\frac{d\varphi}{2\pi}\int_0^{\pi} d\theta \sin\theta \int_0^{\infty}\frac{dk\,k^2}{(2\pi)^2}\,\frac{e^{ikr\cos\theta}}{k^2+\varkappa^2}$$

but $\displaystyle\int_{-1}^{1} d(\cos\theta)\, e^{ikr\cos\theta} = \frac{e^{ikr} - e^{-ikr}}{ikr}$

$$I = \frac{1}{ir}\int_0^{\infty}\frac{dk\,k}{(2\pi)^2}\,\frac{e^{ikr} - e^{-ikr}}{k^2+\varkappa^2} = \text{(change } k \to -k \text{ in the 2}^{\text{nd}} \text{ integral)}$$

$$= \frac{1}{ir}\int_{-\infty}^{\infty}\frac{dk\,k\,e^{ikr}}{(2\pi)^2\,k^2+\varkappa^2} = \frac{1}{4\pi^2 ir}\int_{-\infty}^{\infty} dk\,\frac{k\,e^{ikr}}{(k+i\varkappa)(k-i\varkappa)}$$



$$e^{ikr} = e^{i(k'+ik'')r} = e^{ik'r} \cdot e^{-k''r}$$

$$\Rightarrow I = \frac{1}{4\pi^2 i r} \, 2\pi i \, \mathrm{Res}\left[\frac{k\, e^{ikr}}{(k+i\varkappa)(k-i\varkappa)}\right] =$$

$$= \frac{1}{2\pi r} \, \frac{i\varkappa\, e^{-\varkappa r}}{2i\varkappa} = \frac{1}{4\pi r} \, e^{-\varkappa r}$$

$$\Rightarrow \varphi(\vec{r}, t) = \frac{Q}{4\pi\varepsilon_0} \, \frac{e^{-\varkappa r}}{r}$$

This is Debye screening : $\varphi \sim \frac{e^{-r/r_D}}{r}$

$$r_D = \frac{1}{\varkappa} = \frac{\varepsilon_0 k_B T}{q^2 n}$$

Potential of charge Q
is screened by free charges in plasma
on distances $r \geq r_D = \frac{1}{\varkappa}$.